

Supporting Information for

Title:

Discovery-based approach to identify multiple factors that affect the spin state of coordination complexes using the Evans NMR method

Authors:

Masaki Uchida*, Candice H. Cortney, Karina Bustos, Elia Manzo, Emily Sauls, Jacob Bouchard, Risako Fukazawa, and Viswanathan V. Krishnan

Affiliations:

Department of Chemistry and Biochemistry, California State University, Fresno
2555 E. San Ramon Ave., MS SB70, Fresno, CA 93740, USA

*** Corresponding author: Masaki Uchida**

Email: muchida@mail.fresnostate.edu

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Experimental Procedure

Part 1: Metal acetylacetonate (acac) complexes

Synthesis of $Mn(acac)_3$ ¹

1. Weigh out 1.88 g of $KMnO_4$ in a 150 mL beaker and add it to 38 mL of deionized water.
2. Warm and stir the solution to dissolve the $KMnO_4$ at 80°C.
3. Cool the solution to room temperature in an ice bath.
4. Measure 8.5 mL of acetylacetone and add it slowly to the solution in several aliquots over a few minutes while gently stirring the solution.
5. Boil the mixed solution for 5 minutes while gently stirring the solution, then cool the beaker in an ice bath for about 15 minutes.
6. Collect the precipitate on a coarse-fritted filter.
7. Wash the precipitate with 5 mL of deionized water and repeat three times.
8. Dry the precipitate thoroughly by pulling air with the vacuum.
9. Transfer the precipitate to a clean vial and leave it in a desiccator to dry.
10. After drying, weigh the $Mn(acac)_3$ and calculate the yield.

NMR measurements of metal acac complexes

1. Prepare a clean NMR tube with a coaxial insert (Chemglass, C-NI5CCI-B) for NMR measurement.
2. Add 1 v/v % of $CHCl_3$ and 0.2 v/v % of tetramethylsilane (TMS) to $CDCl_3$.

Note: $CHCl_3$ is supplemented to increase the intensity of $CHCl_3$ 1H NMR peak from the coaxial inner tube. This solution is used to dissolve metal acac complexes. As there were four groups and each group measured five different metal acac complexes, at least 20 mL of this solution was needed. In our class, one group prepared 25 mL solution in a batch for all groups. If time is an issue, this solution could be made in advance by the instructor.

3. Add 0.05 v/v % of TMS to $CDCl_3$.

Note: This solution is added to the outer tube as an internal reference for NMR measurements of metal acac complexes. In our class, one group prepared 15 mL solution in a batch for all groups. If time is an issue, this solution could be made in advance by the instructor.

4. Take approximately 0.007 g of a metal acac complex into a clean vial. Record the weight of the complex to calculate the concentration of the complex.
5. With a micropipette, add 1 mL of CDCl_3 supplemented with 1 v/v % of CHCl_3 and 0.2 v/v % of TMS (prepared in step 2) into the same vial. Shake the vial gently until the metal complex is completely dissolved.
6. Calculate the concentration of the metal acac complex in the sample solution. (It should be about 20 mM).
7. Add approximately 150 μL of the sample solution (prepared at step 5) to the coaxial insert of the NMR tube.
8. Add approximately 600 μL of the reference solution (prepared in step 3) to the outer NMR tube.
9. Place the coaxial insert inside the outer tube until the insert is positioned to full depth within the outer tube (Supporting Figure 3).
10. Acquire the NMR spectrum of the sample solution using a 300 MHz NMR spectrometer (Bruker Fourier 300 spectrometer). Use CDCl_3 for automated locking and shimming. Set the data acquisition for 64 scans and start collecting data.
11. Calibrate the ^1H NMR spectra with TMS in the outer tube on the NMR data.
12. Identify the difference in the observed chemical shift of the reference compounds, TMS and CHCl_3 , between the inner and outer tubes.

Part 2: Metal aqua and metal cyanide complexes

Preparation of $[\text{Co(II)}(\text{H}_2\text{O})_6]^{2+}$ solution²

1. Mix 9.9 mL of D_2O and 0.1 mL of deionized water. Dissolve 0.100 g of sodium 3-(trimethylsilyl)-1-propanesulfonate (DSS) in this solution.
2. Weigh approximately 0.0045 g of $\text{Co}(\text{NO}_3)_2 \cdot 6 \text{H}_2\text{O}$ in a small glass vial. Record the weight of the chemical to calculate the concentration of the complex. Dissolve the chemical by adding 3 mL of D_2O solution made at step 1 to prepare $[\text{Co(II)}(\text{H}_2\text{O})_6]^{2+}$ solution.
3. Calculate the concentration of $[\text{Co(II)}(\text{H}_2\text{O})_6]^{2+}$ in the sample solution. (It should be about 5 mM).

Preparation of $[\text{Co(III)}(\text{H}_2\text{O})_6]^{3+}$ solution^{2,3}

1. Weigh 0.146 g $\text{Co}(\text{NO}_3)_2 \cdot 6 \text{H}_2\text{O}$ into a small glass vial and dissolve the sample in 2.5 mL of D_2O . Add 0.026 mL of 30% H_2O_2 and mix well.
2. In a 25 mL Erlenmeyer Flask, mix 1.00 g of KHCO_3 in 2.5 mL D_2O and 0.026 mL of 30% H_2O_2 . If need be, gently heat this solution on a hot plate to dissolve KHCO_3 . Allow the solution to cool before moving to the next step.
3. With stirring, slowly add the solution from Step 1 to the Erlenmeyer flask from Step 2. Continue stirring for at least 1 minute after completely adding the solution from step 1 to ensure all bubbling stops. Observe the color change.
4. Carefully measure 1.10 mL of concentrated HNO_3 (15.8 M) and add to 3.25 mL of D_2O to prepare a 4 M solution of HNO_3 .
5. Transfer 4 mL of 4 M HNO_3 from step 4 to a clean 25 mL Erlenmeyer flask. Slowly add 1 mL of the solution from step 3 while stirring. Observe the color change. Stir the solution till it no longer bubbles and the color remains constant. (If the solution is brown/black, the synthesis has failed.)
6. Calculate the concentration of $[\text{Co(III)}(\text{H}_2\text{O})_6]^{3+}$ in the sample solution. (It should be about 5 mM).
7. To supplement 1 wt. % DSS, weigh 0.02 g of DSS in a small glass vial. Add 2 mL of the $[\text{Co(III)}(\text{H}_2\text{O})_6]^{3+}$ solution prepared in Step 5 into the vial and dissolve DSS.

Preparation of $[\text{Fe(III)}(\text{H}_2\text{O})_6]^{3+}$ solution⁴

1. Carefully measure 0.43 mL of concentrated perchloric acid (11.6 M) and add 4.57 mL of D_2O to prepare 1 M solution of perchloric acid.

Note: The 1 M perchloric acid solution was prepared by the instructor. The 5 mL solution was enough for four groups.

2. One milliliter of 1 M perchloric acid is diluted with 9 mL of D₂O to prepare 0.1 M perchloric acid solution.
3. To supplement 1 wt. % DSS, 0.100 g of DSS was weighted in a small glass vial and dissolved with 10 mL of 0.1 M perchloric acid solution.
4. Weigh approximately 0.006 g of iron(III) perchlorate in a small glass vial. Dissolve it with 3 mL of 0.1 M perchloric acid solution supplemented with DSS from step 3 to prepare [Fe(III)(H₂O)₆]³⁺ solution.
5. Calculate the concentration of [Fe(III)(H₂O)₆]³⁺ in the sample solution. (It should be about 5 mM).

Preparation of metal cyanide solutions for NMR measurement

1. Mix 9.9 mL of D₂O and 0.1 mL of deionized water. Dissolve 0.100 g of DSS in this solution.
2. Weigh out approximately 0.0085 g of K₄Fe(CN)₆•3H₂O, 0.0066 g of K₃Fe(CN)₆, and 0.0066 g of K₃Co(CN)₆ in a small glass vial, respectively. Add 1 mL of D₂O solution made at step 1 to each vial.
3. Calculate concentration of [Fe(II)CN₆]⁴⁻, [Fe(III)CN₆]³⁻, and [Co(III)CN₆]³⁻ in the sample solutions. (It should be about 20 mM).

NMR measurements of metal aqua and metal cyanide complexes

1. Prepare a clean NMR tube with a coaxial insert (Chemglass, C-NI5CCI-B) for NMR measurement.
2. Dissolve 0.5 wt. % DSS to D₂O.

Note: This solution is added to the outer tube as an internal reference for NMR measurements of metal aqua and metal cyanide complexes. In our class, one group prepared 20 mL solution in a batch for all groups. If time is an issue, this solution could be made in advance by the instructor.

3. Add approximately 150 μL of the sample solution to the coaxial insert of the NMR tube.
4. Add approximately 600 μL of the reference solution (prepared at step 2) to the outer NMR tube.
5. Place the coaxial insert inside the outer tube until the insert is positioned to full depth within the outer tube (Supporting Figure 3).
6. Acquire the NMR spectrum of the sample solution using a 300 MHz NMR spectrometer (Bruker Fourier 300 spectrometer). Acquire the NMR spectrum of the sample solution using

a 300 MHz NMR spectrometer (Bruker Fourier 300 spectrometer). Use D₂O for automated locking and shimming. Set the data acquisition for 64 scans and start collecting data.

7. Calibrate the ¹H NMR spectra with DSS in the outer tube on the NMR data.
8. Identify the difference in the observed chemical shift of the reference compounds, DSS and D₂O, between those in the inner and outer tubes.

Data analysis

1. Determine molar magnetic susceptibility of each coordination complex, χ_M , using equation (1) in the main text.
2. Obtain paramagnetic contribution, χ_M^{para} , using equation (2) in the main text. Diamagnetic contribution, χ_M^{dia} , is obtained using diamagnetic corrections in Table S1.^{1,5}
3. Obtain effective magnetic moment, μ_{eff} , using equation (3) in the main text.
4. Determine the number of unpaired electrons on the metal center of a coordination complex, n, using equation (4) in the main text.

Supplementary Data and Figures

Table S1 Diamagnetic corrections for ligands and metals used for the calculation.^{1,5}

Formula	$\chi_M^{dia}(\text{cm}^3 \text{ mol}^{-1})$
acac ⁻	-52 x 10 ⁻⁶
H ₂ O	-13 x 10 ⁻⁶
CN ⁻	-13 x 10 ⁻⁶
Cr ³⁺	-11 x 10 ⁻⁶
Mn ³⁺	-10 x 10 ⁻⁶
Fe ²⁺	-13 x 10 ⁻⁶
Fe ³⁺	-10 x 10 ⁻⁶
Co ²⁺	-12 x 10 ⁻⁶
Co ³⁺	-10 x 10 ⁻⁶
Ru ³⁺	-23 x 10 ⁻⁶

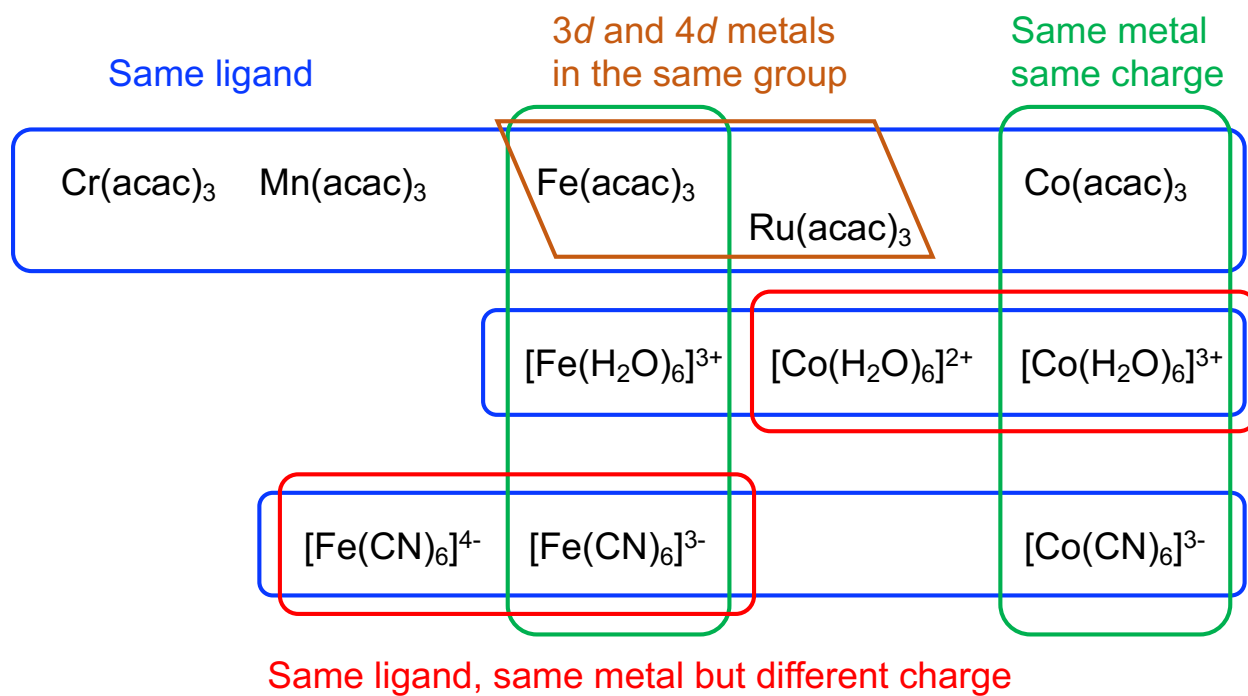
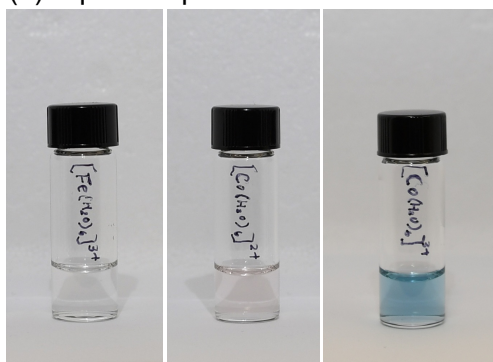


Figure S1 Schematic representation of the design of this laboratory experiment. Through the systematic analysis of the set of coordination complexes, students can investigate various factors that contribute to the *d*-electron configuration and the spin state of the complexes.

(a) Acetylacetonate complexes



(b) Aqua complexes



(c) Cyanide complexes

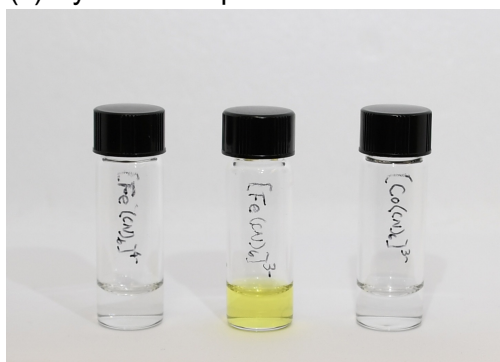


Figure S2 Various coordination complexes examined in this experiment; (a) $\text{Cr}(\text{acac})_3$, $\text{Mn}(\text{acac})_3$, $\text{Fe}(\text{acac})_3$, $\text{Co}(\text{acac})_3$, and $\text{Ru}(\text{acac})_3$ (left to right), (b) $[\text{Fe}(\text{III})(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{II})(\text{H}_2\text{O})_6]^{2+}$, and $[\text{Co}(\text{III})(\text{H}_2\text{O})_6]^{3+}$ (left to right), (c) $[\text{Fe}(\text{II})(\text{CN})_6]^{4-}$, $[\text{Fe}(\text{III})(\text{CN})_6]^{3-}$, and $[\text{Co}(\text{III})(\text{CN})_6]^{3-}$ (left to right).



Figure S3 An example of an NMR tube with sample solution. A solution containing a coordination complex and reference substances is transferred into the coaxial insert, whereas solvent with reference substances is placed in the outer tube.

Instructor's Notes

Experiment timeline

This experiment has been optimized for groups of 3 – 4 students. We had four groups each when it was implemented in Spring 2022 and Spring 2023. The general timeline of this experiment was summarized in Table S2. Students had two separate 3-hour lab sections per week and spent three weeks for this experiment, followed by one additional section for discussion. In terms of synthesizing the complexes ($\text{Mn}(\text{acac})_3$ and three aqua complexes), each group synthesized the complexes for their NMR measurements. In our classes, each group was provided with two set of NMR tube with coaxial insert. Additionally, our NMR has autosampler (Bruker Sample Xpress Lite). Therefore, each group was able to prepare the next NMR sample while one sample was cued for the NMR measurement. It certainly helped to streamline synthesizing four complexes and analyzing 11 complexes in three weeks. If time is limited and the instructor needs to shorten this experiment, it is possible to assign several different complexes to each group instead of having each group measure all 11 complexes. In this case, it would be important to share the NMR data with all groups and have students analyze the whole set of complexes to identify three factors that affect the spin-state of complexes.

Table S2 General timeline of this experiment

Week 1		Week 2		Week 3		Week 4
Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Session 7
Synthesis of $\text{Mn}(\text{acac})_3$ complex	NMR measurement of metal acac complexes	NMR measurement of metal acac complexes	Preparation and measurement of $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	Preparation and measurement of $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$	Measurement of metal cyanide complexes	Discussion of the experimental result

Cleaning of NMR tube

It was important to wash NMR tubes as soon as possible after they were used. Once a solvent in a coaxial inner tube was evaporated and a metal complex was dried out, it was not easy to clean the inner tube well.

If the sample in the inner tube was a metal acac complex, the tube was washed several times with CHCl_3 followed by ethanol after discarding the sample solution into an appropriate waste bottle.

If the sample was a metal aqua or a metal cyanide complex, the tube was washed several times with H₂O followed by ethanol. Tube washing was done in a fume hood. If it was necessary to use the NMR tube again for another measurement in the same lab section, the tube was carefully dried with a heat gun.

Equation to determine magnetic susceptibility by Evans method

The equation derived by Evans to determine the mass magnetic susceptibility of the solute χ_g (cm³/g) for NMR with permanent magnet or electromagnet is as follows:⁶

$$\chi_g = \frac{3\Delta\nu}{2\pi m\nu_0} \quad (S1)$$

where m is the mass of the solute in g/cm³, $\Delta\nu$ is the observed peak shift, ν_0 is the frequency of the NMR spectrometer in hertz. For NMR with superconducting magnet, this equation was modified as:^{7,8}

$$\chi_g = \frac{-3\Delta\nu}{4\pi m\nu_0} \quad (S2)$$

On the other hand, some other articles use equations with opposite sign.^{9,10} Namely, the following equation:

$$\chi_g = \frac{-3\Delta\nu}{2\pi m\nu_0} \quad (S3)$$

for NMR with permanent magnet or electromagnet and the following equation:

$$\chi_g = \frac{3\Delta\nu}{4\pi m\nu_0} \quad (S4)$$

for NMR with superconducting magnet.

The important point of these equations, in terms of peak shift difference between conventional NMR with permanent magnets or electromagnets and high-field NMR with superconducting magnet, is that the peak shift of a reference signal in the presence of a paramagnetic solute observed with high-field NMR is “twice in magnitude (in ppm) and of the opposite sign from that expected in the case of conventional spectrometers” as described in Live and Chan’s paper.⁷ We think that the difference in sign between equations S1, S2 and S3, S4 comes from whether the authors define positive $\Delta\nu$ as NMR peak shift to upfield or downfield. In our experience, most students consider downfield peak shift due to a paramagnetic complex as a positive shift (i.e., positive $\Delta\nu$), probably because the chemical shift, δ , increases. Thus, we think that equations S3 and S4 are appropriate to avoid confusion of students. The equation 1 in the main manuscript is converted from equations S3 and S4 for determining molar magnetic susceptibility χ_M .

It should be noted two terms (solvent correction terms) have been removed from equation S1 – S4 and equation 1, because these terms almost cancel each other out.^{9, 11}

Ligand exchange in aqua complexes

Exchange a coordinated water molecules with a free (solvent) water molecule would affect the chemical shift of water signal observed in NMR spectra. Indeed, in the case of aqua complexes, there was around 10 -20 % difference in the number of unpaired electrons determined by the two reference compounds (DSS and H₂O), while the difference was less for acetylacetonate complexes and cyanide complexes. Nevertheless, the number of unpaired electrons obtained from the two reference compounds were close enough to rationally identify the spin-states of the aqua complexes. Thus, we think using the two reference compounds is useful as it gives students confidence in their results. The difference in the number of unpaired electrons determined from DSS and H₂O would provide an interesting discussion for advanced students on the effect of ligand exchange on NMR peak shifts.

Hazards

In our institute, chemical waste was typically collected separately in four different categories: 1) aqueous waste, 2) non-halogenated organic waste, 3) halogenated organic waste, and 4) solid waste. In this experiment, the cyanide complexes were collected in a separate waste container, besides the four waste containers. We also collected the unused diluted perchloric acid solution separately.

Table S3 List of chemicals used for this experiment

Chemical	CAS number
Potassium permanganate	7722-64-7
Acetylacetone	123-54-6
Chromium(III) acetylacetonate	21679-31-2
Iron(III) acetylacetonate	14024-18-1
Cobalt(III) acetylacetonate	21679-46-9
Ruthenium(III) acetylacetonate	14284-93-6
Cobalt(II) nitrate hexahydrate	10026-22-9
Potassium hydrogen carbonate	298-14-6
Nitric acid	7697-37-2
Iron perchlorate hydrate	15201-61-3
Perchloric acid	7601-90-3
Potassium ferrocyanide trihydrate	14459-95-1
Potassium ferricyanide	13746-66-2
Potassium hexacyanocobaltate	13963-58-1
Chloroform	67-66-3
Chloroform-d	865-49-6
Deuterium oxide	7789-20-0
Tetramethylsilane (TMS)	75-76-3
Sodium trimethylsilylpropanesulfonate (DSS)	2039-96-5

Table S4 List of supplies and equipment used for this experiment

Items	Per group	Note
<i>For synthesis of Mn(acac)₃</i>		
150 mL beaker	1	Synthesis of Mn(acac) ₃
50 mL graduated cylinder	1	
10 mL graduated cylinder	1	
Coarse fritted filter	1	
500 mL side arm flask	2	For vacuum filtration
Stand with clamps	1	
Filter adapter	1	
Vacuum tube	2	
Vacuum pump	1	Water aspirator works as well
Stirring hot plate	1	
Ice bucket	1	
Thermometer	1	
Magnetic stir bar	1	
9" Pasteur pipets	3	
Spatula	1	
Glass rod	1	
Rubber policeman	1	
20 – 30 mL glass vial	1	For storage of Mn(acac) ₃
Desiccator	1 (per class)	
<i>For preparation of [Co(II)(H₂O)₆]²⁺</i>		
10 mL graduated cylinder	1	
4 mL glass vial	1	
200 µL pipette	1	
Spatula	1	

Table S4 List of supplies and equipment used for this experiment (continued)

Items	Per group	Note
<i>For preparation of $[\text{Co(III)}(\text{H}_2\text{O})_6]^{3+}$</i>		
25 mL Erlenmeyer flask	2	
10 mL graduated cylinder	1	
4 mL glass vial	1	
10 mL glass vial	1	
1 mL pipette or syringe	1	
200 μL pipette	1	
Spatula	1	
Stirring hot plate	1	
Small magnetic stir bar	2	For 25 mL Erlenmeyer flask
<i>For preparation of $[\text{Fe(III)}(\text{H}_2\text{O})_6]^{3+}$</i>		
10 mL graduated cylinder	1	
25 mL glass vial	2	
4 mL glass vial	1	
1 mL pipette or syringe	1	
Spatula	1	
<i>For NMR measurement</i>		
NMR tube with a coaxial insert	2	Chemglass catalog number C-NI5CCI-B (for Bruker NMR) or C-NI5CCI-V (for Varian NMR)
25 mL glass vial	2	
4 mL glass vial	10	To dissolve complexes
1 mL pipette or syringe	1	
200 μL pipette	1	
20 μL pipette	1	
Spatula	1	

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